Feature Fusion Network Model Based on Dual Attention Mechanism for Hyperspectral Image Classification

Ying Cui, WenShan Li, Liwei Chen, Liguo Wang, *Member, IEEE*, Jing Jiang, and Shan Gao

***Abstract*— Hyperspectral images (HSIs) have been playing an important role in the field of ground object classification because of their rich spatial and spectral information. Aiming at how to extract complex feature information from HSIs, we propose a new feature fusion network based on dual attention (DAFFN) with dual attention mechanism, which is mainly used to capture more accurate global–local context attention features. The model extracts global context attention features using self-attention (SA) mechanism and local context attention features using cross-attention mechanism. Considering the problem that position information is easily lost during the conversion of attention mechanism, we propose a position self-calibration (PSC) module that can be flexibly embedded into two attention modules. In addition, to better integrate global and local features, we also designed a multiscale global and local feature fusion module (MSGL), which preserves more representative features with less communication costs by aggregating global and local attention features. We have carried out experiments on three commonly used hyperspectral datasets, and the classification results show that our model can achieve high classification accuracy even in the case of a limited number of samples.**

***Index Terms*— Attention mechanism, feature fusion, global context feature, hyperspectral image (HSI), local context feature, position calibration.**

I. Introduction

HYPERSPECTRAL sensors collect hundreds of narrow-band and continuous spectral bands, so they can provide rich information for the classification and recognition of targets in satellite scenes. Through the processing of hyperspectral image (HSI), each pixel of the image can be accurately analyzed and classified. At present, HSI classification technology has been widely used in environmental monitoring [1], geological exploration [2], crop analysis [3], military [4], and other fields.

In recent years, the classification model based on convolutional neural networks (CNN) has become a representative classification method in the field of HSI classification, because of its unique hierarchical feature learning ability and automatic feature learning ability. Chen et al. [5] with HSI cube as input, the whole sample is represented by the central pixel category. After training by CNN, the classification accuracy is improved, and a new method of sample processing is provided for HSI classification. Yu et al. [6] apply CNN to the classification of HSI. Different from the traditional classification method, an end-to-end automatic learning feature method is proposed. A pixel and its neighborhood in HSI are used as the input of CNN, and the output result is the predicted category label. Lee and Kwon [7] convolved HSI with different convolution kernel sizes, explored local context interaction information, and obtained multiscale spatial–spectral feature map. However, the referenced 2-D-CNN network also extracted some noise features during HSI convolution, resulting in classification errors. Yang et al. [8] gradually reduce the central pixel in the patch set by recursion to extract features, which effectively solves the problem that 2-D-CNN and 3-D-CNN models introduce unnecessary noise while extracting spatial information. Wang et al. [9] designed two subnetworks to extract spectral and spatial feature information, respectively. Through feature stitching and fractional weighting, adaptive feature fusion was carried out to avoid the problem of not obvious features extracted by shallow network and over-fitting caused by deep network. To extract more in-depth HSI features, a deeper and more complex learning model structure was proposed. Song et al. [10] introduced a residual network to fuse features at different levels, extract more discriminative features, and avoid the problem of extraction gradient disappearance and gradient explosion caused by the increase of network layers. Zhong et al. [11] proposed a spectral spatial residual network (SSRN), which alleviates the problem of accuracy degradation through residual connections and 3-D-CNN. Wang et al. [12] extracted spectral and spatial features using different convolutional kernel sizes, and extracted deep features using dense

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connected structures. Zhang et al. [13] aiming at the problem of losing fine features during deep learning, a multiscale dense network (MSDN) was proposed to realize the reconstruction of depth feature extraction and multiscale feature information fusion. Song et al. [14] extracted spectral spatial features using the 3-D multiscale DenseNet method; it can share the features of different blocks, enhance the information flow, solve the problem of insufficient training samples, and alleviate the problem of gradient disappearance. Pan et al. [15] use the expansion convolution strategy to avoid the impact of resolution reduction, and aggregate multiscale context information from HSI data. To reduce the network complexity and execution cost, Wang et al. [16] proposed a HyperLiteNet model, which is a lightweight nondeep parallel network, achieving high efficiency and high performance at a lower computational cost. Li et al. [17] proposed a deep spectral feature fusion method, which extracts the spectral features between local and local, and also captures the correlation between nonadjacent bands, making up for the shortcomings of spectral–spatial feature fusion. Cui et al. [18] proposed a multiscale dense and self-calibrated solution network (MDSCN) to extract local context features in spatial and spectral dimensions, respectively. Ding et al. [19] constructed a new multiscale graph sample and aggregation network using context aware mechanisms to characterize the importance between adjacent spatial regions, and used context aware learning methods for HSI classification. Wang et al. [20] observed that there are many homogeneous regions with different semantic and geometric attributes in HSI, which can be used to extract useful contexts. They designed a clustering scheme for the generation of Contour line adaptive homogeneous regions, and used transformers to capture the regional adaptive and global aggregation spatial background located, respectively, between these regions or regions. Although CNN-based models can achieve good classification results, it still faces some problems such as high processing time and computational complexity, large training sample requirements, and overfitting. And most CNN-based methods struggle to efficiently preserve the details of irregular cross-class edges and small irregular regions, and rarely make full use of the local neighborhood environment, resulting in the loss of context detail feature information of HSI.

Recently, researchers introduce attention mechanism into HSI classification model, which is designed to focus more on the informative areas and takes less account of nonessential areas. Ma et al. [21] proposed a dual band multiattention (DBMA) mechanism network that utilizes convolutional block attention modules to integrate important spectral and spatial information. Later, Li et al. [22] proposed a double-branch dual-attention (DBDA) network based on the DBMA. The overall architecture of the network is consistent with DBMA, but the composition of the attention module and the activation function are different from DBMA. Hang et al. [23] proposed a two-branch spectrum–spatial attention network, which enables the neural network to focus on more different channels and spatial positions while suppressing unnecessary channels and spatial positions. The above networks are all based on the two-branch attention architecture for hyperspectral classification, but the attention modules of two branches are separated from each other. Based on this problem, Cui et al. [24] introduced the effective channel attachment (ECA) module in DenseNet, built a cross-dimensional interaction model between channel dimension and spatial dimension, and realized the high-precision classification of HSI on the basis of capturing cross-dimensional interaction information. To reduce the problem of redundant computation in the spatial attention model, Cui et al. [25] proposed a lightweight classification model based on the spatial–spectral attention mechanism, which can achieve high classification accuracy and reduce the complexity. Zhang et al. [26] proposed a new global–local block spatial–spectral fusion attention model, which uses the self-attention (SA) mechanism and spatial pyramid pool (SPP) to focus on the edge feature information of local blocks. A pyramid conditional convolution attention network, which fully captures the information of hyperspectral input data through convolution kernels with different weights and scales [27]. Wang and Fan [28] proposed the attention mechanism module to extract the weight information of spatial dimension, spectral dimension, and channel dimension, respectively, and fuse the weight information of the three dimensions through feature multiplication, and embed it into the end of each scale channel. Xia et al. [29] combined the SA mechanism and the residual structure, the input features and output features can be adaptively fused, further enhancing the feature extraction ability of the residual structure. Xu et al. [30] proposed multiscale and cross hierarchical attention learning networks (MCAL) to fully explore the global local context multiscale features of HSI, addressing the drawback of the transformer model not being able to model local information. Lin et al. [31] proposed a context aware attention graph U-Net to enhance contextual features within and between classes, focusing on the correlation between intraclass and interclass features. In the aforementioned attention-based methods for HSI classification, most studies directly embed the SA mechanism into the transformer, although it is hoped to contribute to the acquisition of global information in HSI. However, as the layers become deeper, the network complexity increases, leading to the transformer model suffering from network degradation. And some studies have also introduced SA mechanisms into pixel level CNN to obtain spatial spectral information of HSI separately. Although some results can be achieved, they have not detailed feature extraction into global and local levels, thus failing to achieve effective discriminative feature extraction.

The previous methods that combine positional embedding are mostly based on transformers and do not have portability to any model, and the embedded position also has an impact on the performance of the model.

In view of the above problems, this article proposes a feature fusion network based on dual attention (DAFFN) based on the dual attention mechanism. First, the model processes the global–local context interaction information and spatial dependence information through the SA [32] and cross-attention [33] modules to obtain more accurate global–local context features. The SA mechanism is based on the attention to the self-characteristics of HSI, and the cross-attention mechanism is based on the attention to the

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features on the pixel crossing path. Second, to solve the problem of position information extraction in hyperspectral data, we propose a position self-calibration (PSC) module which can be embedded in any attention mechanism. This module calibrates the convolution transformation of other convolution layers through low-dimensional embedding, which enables the pixels at each position to adaptively encode the information at a long distance. To improve the parameter efficiency, we expand the channel width of key modules of the cross-attention mechanism and reduce the channel width of unimportant modules to redistribute the parameters in the network. Using this strategy, important modules have more channels to learn representations in high-dimensional space, avoiding the loss of feature information. Third, to better fuse the features extracted by the two attention modules, we propose a multiscale global–local feature fusion module (MSGL), which extracts more discriminative hierarchical features from the information encoded by the attention module through the convolution layers of different scales, and better fuses the context features at the pixel level. We propose that the model DAFFN is composed of a global feature extraction branch composed of a SA position calibration module (SA-PSC) and a local feature extraction branch composed of a cross-attention position calibration module. The feature fusion module shares the acquired global–local context features. DAFFN can achieve high-precision classification with limited training samples and less training time and computation.

Our innovative contributions can be summarized as follows.

1) The SA mechanism and context coding are introduced to extract global context information from HSI. The main goal is to connect each element in the feature map, thereby improving access to all pixels' receptive field and obtaining global context features.
2) The cross-attention module is introduced to obtain the details of HSI, and forms a complementary model with the SA module, so that the model can focus on the local context features of HSI.
3) A PSC module that can be embedded in any attention mechanism is proposed. Embedding the module into the attention module can realize the adaptive encoding of long-range information and calibrate additional context information to extract more fine-grained global–local feature.
4) To better fuse the global and local context features, an MSGL is proposed. By connecting all the hierarchical features of the two channels together and utilizing the complementarity between features, the effective features are retained to the maximum extent.

We have carried out experiments on three HSI datasets, and the experimental results show that. The proposed method achieves excellent classification performance with low amount of parameters and computation. The rest of this article is organized as follows. Section II introduces the proposed method and model. Section III, a series of experiments are carried out to verify the effectiveness of the proposed method, and the experimental results are analyzed. Section IV, summarizes the full text.

II. Method

In this section, we will introduce the proposed method for HSI classification in detail. SA module, cross-attention module, and feature fusion module proposed in this article are described, respectively, for extracting global–local context features.

A. Model Structure Framework

As shown in Fig. 1, we preprocess the HSI, assuming that the input image is $X \in R^{B\times W\times H}$ where H and W are the height and width of the spatial dimension, and B is the spectral dimension of HSI. Considering that HSI contains hundreds of spectral channels, there is a large amount of redundant information, which can easily cause the "Hughes" phenomenon if processed simultaneously. Therefore, this article adopts principal component analysis (PCA) [34] dimensionality reduction, which can reduce the spectral dimension and preserve the effective spectral feature information of HSI while keeping the spatial dimension unchanged. First, we use PCA dimension reduction to solve the problem of high HSI dimension and correlation and redundancy between different bands, and get $X \in R^{C\times W\times H}$, C is the number of spectral bands after dimensionality reduction. Next, to control the parameter quantity of the model, we use deep separable convolution to obtain the spatial and spectral feature maps of HSI. Deep separable convolution achieves channel and spatial separation, which can reduce the required parameters while maintaining the performance of standard convolution. To obtain effective global context attention features, we send the initial feature map into the SA position calibration module to enhance the feature information; at the same time, the initial feature map is also sent to the cross-attention position calibration module to obtain local effective information. To better fuse features with inconsistent semantics and scales, we propose an MSGL, which can be used to obtain both lower level features with high resolution and higher level features with rich semantics. Then, the Focal Loss function is used to replace the cross entropy loss function to solve the problem of low classification accuracy caused by sample imbalance; Finally, get the label of each pixel, and then get the final HSI classification map.

B. Self-Attention Position Calibration Module

Due to the small number of HSI samples, high feature redundancy, and a large amount of spatial and spectral semantic information, this information has a great impact on the classification effect of the classification model. If too many redundant features are extracted, it will not only lead to recognition errors of ground objects, but also increase additional computing costs. Therefore, we expect the proposed network to see global features while also focusing on key information. When processing HSI image data, each pixel in the feature map is often contextual, but the correlation with different pixels in the context is different. Therefore, when processing this pixel, we need to focus more on the pixels with higher correlation while seeing its context. We propose a new SA-PSC to address the above issues. Next, we introduced SA and the PSC module.

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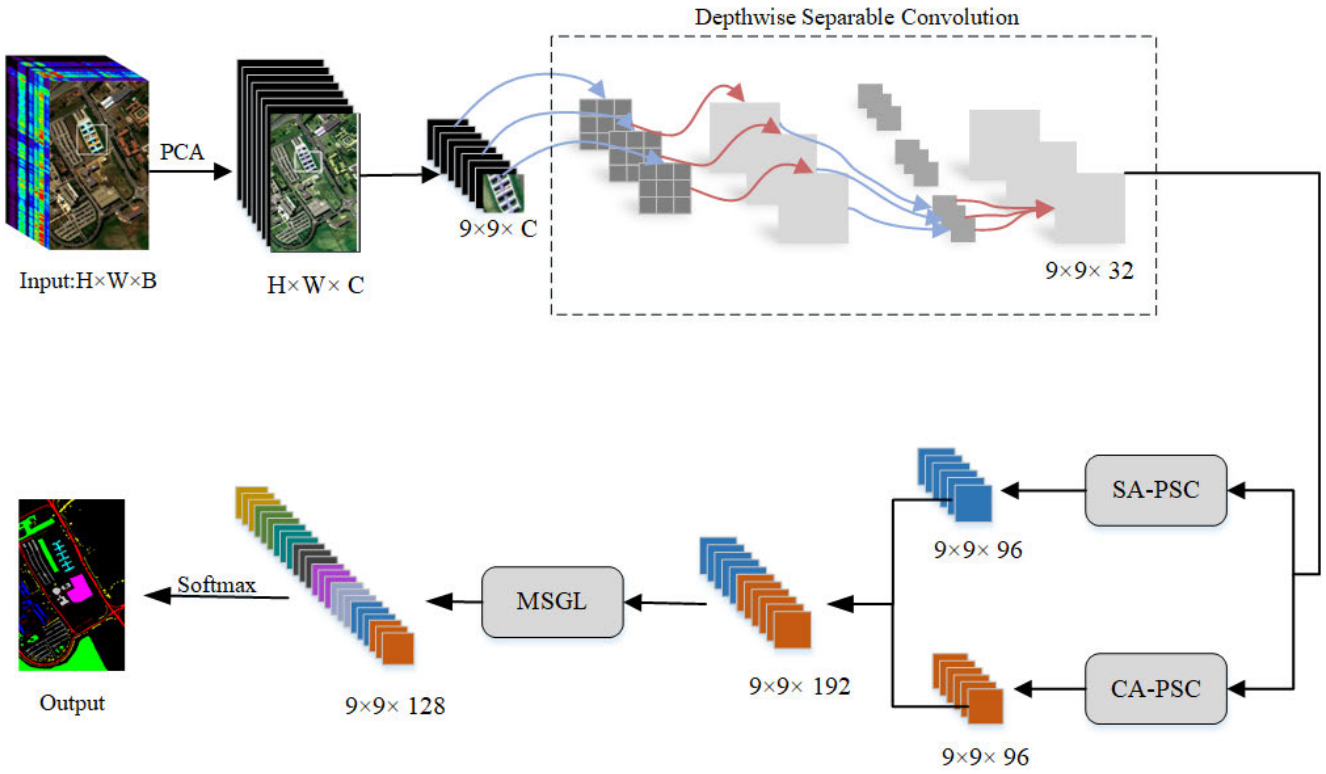


Fig. 1. Structure of feature fusion network model with dual attention mechanism. The original HSI is reduced to the specified dimension after PCA processing, and softmax function is an activation function for multiclass classification problems.

The SA module utilizes the global interaction between the semantic features at the end of the encoder to explicitly model complete contextual information, aiming to extract remote structural information from HSI. The SA module connects each pixel in the top-level feature mapping, finds the input features that deserve more attention, and finally outputs the receptive field containing all the effective context information. Therefore, we use the SA module to construct global dependence relationships between spectral and spatial domains. As shown in the SA module in Fig. 2, the SA module has three inputs representing embedding of different learning features, namely the query matrix Q, key matrix K, and value matrix V. The embedding matrix is denoted as W_q, W_k, and W_v

$$Q = W^q X \tag{1}$$

$$K = W^k X \tag{2}$$

$$V = W^v X \tag{3}$$

where X is the input vector matrix, W^q, W^k, and W^v are embedded matrices, and the output attention matrix is

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V = AV \tag{4}$$

where A is the attention matrix, and each row of A corresponds to the similarity between the given element in Q and all elements in K. Then, the attention function performs a weighted average of the elements with a value of V to account for all interactions between the query and the key.

Note that one row of the matrix Attention(Q, K, V) corresponds to the similarity between the given pixel in Q and all pixels in K. The attention function weights the pixels of V to explain all interactions between Q and K. Three 1×1 convolutions are defined in the initialization function, namely query_conv, key_conv, and value_conv, to learn different types of features of HSI and obtain Q matrix, K matrix, and V matrix. Next, to achieve a direct dependence relationship between any two elements in the global context, we reconstruct the Q, K, and V matrices and multiply the transposions of Q and K to obtain the feature maps. Then, the rows of the feature map are normalized using the softmax function. After normalization, the sum of each row is 1 to obtain the weight of each pixel relative to other pixel positions. Finally, the attention feature map is multiplied by the value matrix V to obtain the global context attention enhancement feature X_1.

However, the SA mechanism does not have the function of recording position information, which can easily cause disorder of feature sequences in feature fusion. Therefore, we want to enable it to perform PSC without changing its own SA mechanism. Inspired by the self-calibrated convolution proposed in [35], we designed a PSC module. The proposed PSC module is different from the existing position coding applications in attention mechanisms in the following aspects. First, the application of existing position coding in the attention mechanism often needs to be implemented based on transformer model and requires additional coding and decoding steps and learnable parameters, which consumes a lot of computation. However, the PSC module proposed by us encodes based on 2-D information and can be directly applied to CNN. It no longer needs to be bound with the

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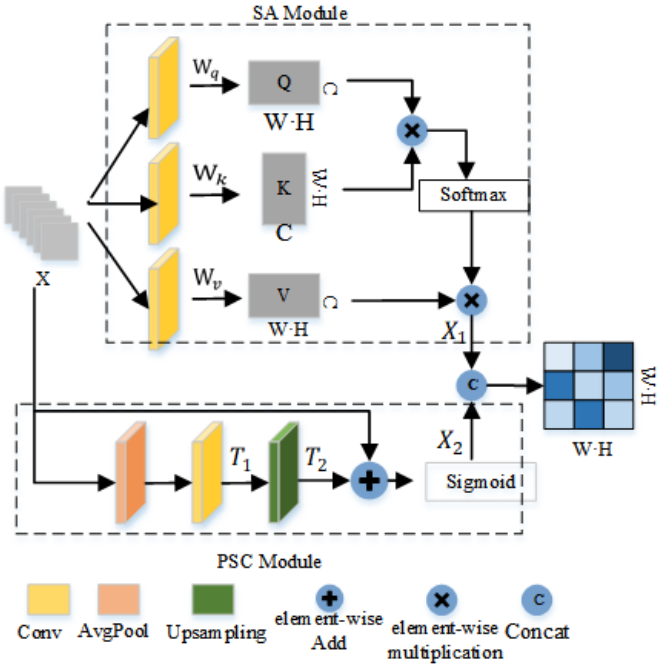


Fig. 2. Structure of SA-PSC. X is the input feature map obtained through deep separable convolution.

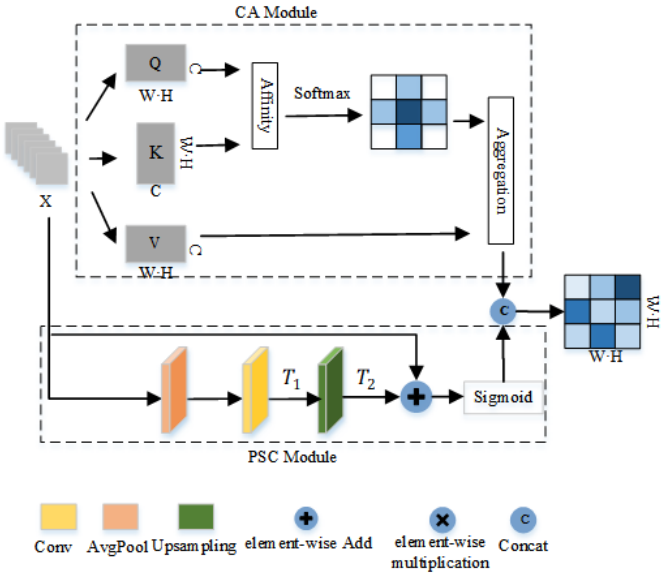


Fig. 3. Structure of CA-PSC. Affinity function used to generate local attention. Aggregation is the aggregation context feature function.

transformer model and can be embedded anywhere in other models, making it flexible and adaptive and eliminating the need for additional learnable parameters. Second, currently, the SA mechanisms with position encoding are based on transformer models, and converting to CNN requires additional encoding and decoding steps. Our proposed PSC is based on 2-D information for encoding, which can be directly applied to CNN and embedded in any position of the model. The PSC module first sets up an arbitrarily sized pooling kernel and encodes the position of each channel within a low-dimensional spatial range, allowing each spatial location not only to adaptively treat its surrounding information environment as embedding from low-dimensional space, but also to use this as a scalar for attention mechanisms. Then, perform a 1×1 convolution transformation on the matrix output from the pooling layer, which can model the dependencies between channels. Then it is up sampled to align the spatial position of the attention module, to expand the field of view of each convolution layer, and adaptively establish a long metric space calibration operation at each position. Through the position self-tuning mechanism, the network can accurately calibrate the characteristics of the corresponding target position under the global receptive field, and combine with the calibrated module to obtain more abundant information to generate more identifiable representations.

As shown in Fig. 2, we have built an SA-PSC based on the characteristics of HSI. The PSC module is composed of two branches. The first branch preserves the original feature information for the original feature map. The second branch uses the average pooled downsampling of the input features to reduce the input size and expand the receptive field to obtain the feature map, then uses the 3×3 convolution layer to model the channel relationship, and then uses bilinear interpolation to upsampling the feature map to the size of $(32, 32, 9, 9)$, and uses the sigmoid activation function to calibrate the extracted features of the attention module to obtain the output feature X_2

$$T_2 = \mathrm{Up}(\mathcal{F}_2(T_1)) = \mathrm{Up}(T_1 * K_2) \tag{5}$$

where Up() is a bilinear interpolation operator, $\mathcal{F}$ is a convolution operation, and $*$ is a convolution operation symbol. Finally, a weighted connection is made between the SA module and the position calibration module to obtain the feature map after the position calibration Y_1

$$Y_1 = \sigma_1 X_1 + \sigma_2 X_2 \tag{6}$$

where σ_1 is the weight of the SA module and σ_2 is the weight corresponding to the self-calibration module.

C. Cross Attention Position Self- Calibration (CA-PSC)

The description based on global information often ignores the influence of context Semantic information in the pixel domain, and lacks semantic rich details. To generate dense pixel level context information, we introduce cross-attention module (CA). CA can more effectively obtain local contextual attention features from remote dependencies. CA closes irrelevant or noisy areas from the skip connection function and highlights areas of interest to the application. Local features first collect contextual information in both horizontal and vertical directions through a cross-attention module. Then, by sending the feature map generated by the first cross-attention module to the second cross-attention module, the additional contextual information obtained from the cross path ultimately achieves complete image dependence for all positions.

Since the cross-attention mechanism has no function of recording position information, we added the PSC module, whose structure is the same as that described in Section II-B.

Fig. 3 shows the cross-attention position calibration module. Based on the attention to the high-level feature graph, the

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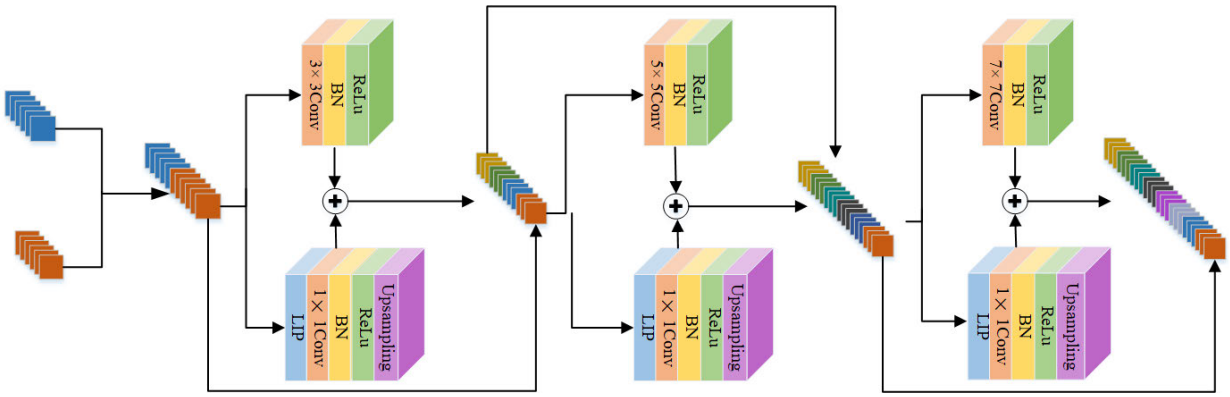


Fig. 4. Structure of MSGL. The input feature map shows global contextual features in blue and local contextual features in orange. ⊕ means elementwise Add.

module is designed as a strobe operation of jump connection. First, the same as the SA module, the query matrix Q, the key matrix K and the value matrix V are obtained through three 1×1 convolution operations, where $\{Q, K\} \in R^{C\times W\times H}$. Then the Q, K, V matrices are reconstructed, and the Q and K matrices are sent into the affinity function to generate a local attention graph. At each position "u" on the spatial dimension of the feature graph Q, we can get the vector $Q_u \in R^C$. Similarly, we can obtain the set Ω_u by extracting feature vectors from K. K is in the same row or column as u. So $\Omega_{i,u} \in R^{(H+W-1)\times C}$, $\Omega_{i,u} \in R^C$ is the ith element of Ω_u.

Then, through sigmoid activation function, the calculated weight value is rescaled between 0 and 1, and the resulting tensor is a filter, where the low-amplitude element represents the noise or uncorrelated area to be reduced. Combine with the repurchase matrix obtained from the value matrix V, and use the aggregation function to obtain the context feature attention graph

$$d_{i,u} = Q_u \Omega_{i,u}^T \tag{7}$$

where $d_{i,u} \in D$ represents characteristic Q_u and characteristic $\Omega_{i,u}$ the degree of correlation and satisfy $i \in [1, \ldots, (H+W-1)]$. It can be seen from this that the data amount attention mechanism is actually for the 1-D vector representation of two pixel points, rather than individual pixel points. Their correlation calculation adopts the extremely simple vector product operation. In addition, $D \in R^{[H+W-1][H\times W]}$, $d_{i,u}$ is in a 2-D tensor plane, and i represents the channel dimension, and u represents the spatial

$$H'_u = \sum_{i\in|\phi_u|} A_{i,u}\phi_{i,u} + H_u \tag{8}$$

where H'_u represents the feature vector of the output feature map $H'_u \in R^{C\times H\times W}$ at position u. $A_{i,u}$ is a scalar value at channel i and location u. Add context information to local feature H to enhance the local feature and the representation by pixel. Therefore, it has a broad context view and selectively aggregates the context according to the spatial attention map.

As shown in Fig. 3 the PSC module is the same as in Section II-B. Finally, the cross-attention module and the position calibration module are weighted and connected to obtain the output features.

D. Feature Fusion Module

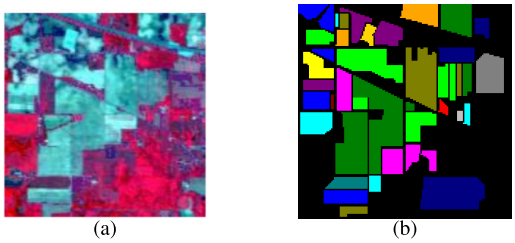
Feature fusion method is an important method in the field of pattern recognition. It can make comprehensive use of multiple image features, realize the complementary advantages of multiple features, and obtain more robust and accurate recognition results. In the standard classification model, feature fusion is usually achieved by directly adding different levels of feature maps, ignoring the differences between different levels of feature maps. We propose an MSGL based on the difference between the global context attention feature extracted by the SA mechanism and the local context attention feature extracted by the cross-attention mechanism.

The MSGL module integrates features from different levels, as shown in Fig. 4. When processing the global feature section, we use convolutional kernels of different scales to decompose the semantic features of different levels of HSI. For the first time, the convolutional layer uses convolutional kernels of 3×3, stride size of 1, and padding of 1 to obtain the global multiscale deep feature map. For the second time, a convolutional layer with a kernel of 5×5, a stride size of 1, and a padding of 2 was adopted; The third time we used a convolutional layer with a kernel of 7×7, a stride size of 1, and a padding of 3. To maintain stable convergence effect and prevent overfitting during network training, batch normalization (BN) layer and activation function rectified linear unit (ReLu) are added after the convolution layer. The lower half of MSGL processes local attention features. To avoid damaging the local context feature information of the CA-PSC module, we first use LIP based on local importance modeling for pooling operations. The size of the pooling layer's core is 3×3, which can adaptively learn the importance weight of each feature and preserve the local importance information of each pixel's neighborhood. Then we add 1×1 convolutional layers to enhance learning ability and ensure smooth features. Next, BN layer and activation function ReLu and upsampling are added. Among them, the upsampling layer is used to adjust the feature map to the same size as the global feature map above for convenient feature fusion. Next, we use dimension addition to aggregate the above two feature maps, and use residual structural join to connect deep and shallow features. Finally, after three feature fusion processes, a feature map was obtained.

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TABLE I
NUMBER OF SAMPLES OF THE IP DATASET

Color	Land-cover-type	Samples
	Alfalfa	46
	Corn-notill	1428
	Corn-mintill	830
	Corn	237
	Grass-Pasture	483
	Grass-Trees	730
	Grass-Pasture-Mowed	28
	Hay-Windrowed	478
	Oats	20
	Soybean-notill	972
	Soybean-mintill	2455
	Soybean-cleantill	593
	Wheats	205
	Woods	1265
	Buildings-Grass-Trees	386
	Stone-Steel-Towers	93
	Total samples	21025



(a) (b)

Fig. 5. (a) Color image of IP dataset. (b) Label map of IP dataset.

III. EXPERIMENTAL RESULTS AND ANALYSIS

In this section, we carry out experimental analysis of the proposed method. First, we introduce four standard HSI datasets used in this article, namely the University of Pavia(UP), Indian Pines (IP), Salinas Valley (SV) and Houston (HT). Next, we compare the classification results of DAFFN and a series of most advanced models on the above three datasets to verify the effectiveness of DAFFN. The experimental results show that DAFFN has certain advantages in classification accuracy and feature extraction.

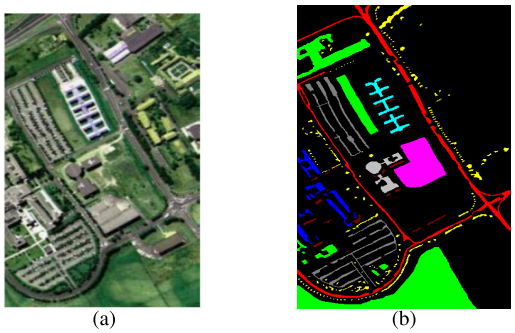
A. Hyperspectral Datasets

1) Indian Pines Dataset: IP dataset is hyperspectral data collected by AVIRIS [36] infrared imaging spectrometer in the IP experimental area in 1992. There are 224 bands in the IP dataset. To reduce the impact of noise bands, 24 water absorption bands are removed and the remaining 200 bands are tested normally. Its wavelength range is 0.4–2.5 μm. 145 $\times$ 145 pixels, with a geometric resolution of 20 m/pixel (m/p). The IP dataset contains 16 kinds of features, with a total of 10 249 samples. Table I lists the name of each category and the corresponding label color and sample number. Fig. 5 shows the pseudo-color image and real label image of IP.

2) University of Pavia Dataset: UP dataset is hyperspectral data collected by ROSIS [37] spectral imager on UP in Italy in 2003. The UP dataset consists of 115 bands in total, 12 unavailable bands are removed, and 103 effective bands are used, with a wavelength range of 0.43–0.86 μm. By 610 $\times$ 340 pixels, with a geometric resolution of 1.3 m/p. The UP dataset contains nine kinds of features, with a total of 42 776 samples. Table II lists the name of each category and the corresponding label color and sample number. Fig. 6 shows the pseudo-color image and the real label image of UP.

TABLE II
NUMBER OF SAMPLES OF THE UP DATASET

Color	Land-cover-type	Samples
	Asphalt	6631
	Meadows	18649
	Gravel	2099
	Trees	3064
	Painted metal sheets	1345
	Bare Soil	5029
	Bitumen	1330
	Self-Blocking Bricks	3682
	Shadows	947
	Total samples	207400



(a) (b)

Fig. 6. (a) Color image of UP dataset. (b) Label map of UP dataset.

3) Salinas Valley Dataset: SV dataset was collected and sent hyperspectral data by AVIRIS imaging spectrometer to SV, CA, USA in 1998. The SV dataset consists of 224 bands, 20 noise and water absorption bands are removed, and 204 effective bands are used. SV dataset consists of 512 $\times$ 217 pixels, with a geometric resolution of 3.7 m/p. The SV dataset contains 16 kinds of features, with a total of 54 129 samples. Table III lists the name of each category and the corresponding label color and sample number. Fig. 7 shows the pseudo-color image and real label image of SV.

4) Houston Dataset: The Houston (HT) dataset was obtained by the ITERS CASI-1500 sensor in rural areas around HT, TX, USA, with a spatial resolution of 2.5 m and a data size of 349 $\times$ 1905 contains 144 bands, with a band range of 364–1046 nm. The HT dataset contains 15 types of features, with a total of 15 029 samples. Table IV lists the names of each category, their corresponding label colors, and the number of samples. Fig. 8 shows HT's pseudo color image and real label image.

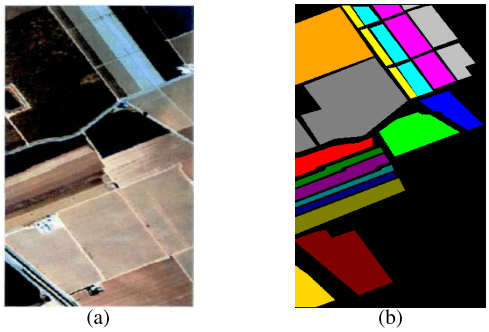
B. Experimental Parameter Setting

The computer equipment used in the experiment was configured on a computer with Intel Core i7-10700 CPU, 2.90 GHz, 64-GB RAM, and NVIDIA GeForce RTX3060. Our in-depth learning framework is built on Python 1.7.1.

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TABLE III
NUMBER OF SAMPLES OF THE SV DATASET

Color	Land-cover-type	Samples
	Brocoli_green_weeds_1	2009
	Brocoli_green_weeds_2-notill	3726
	Fallow	1976
	Fallow_rough_plow	1394
	Fallow_smooth	2678
	Stubble	3959
	Celery	3579
	Grapes_untrained	11271
	Soil_vinyard_develop	6203
	Corn_senesced_green_weeds	3278
	Lettuce_romaine_4wk	1068
	Lettuce_romaine_5wk	1927
	Lettuce_romaine_6wk	916
	Lettuce_romaine_7wk	1070
	Vinyard_untrained	7268
	Vinyard_vertical_trellis	1807
	Total samples	111104



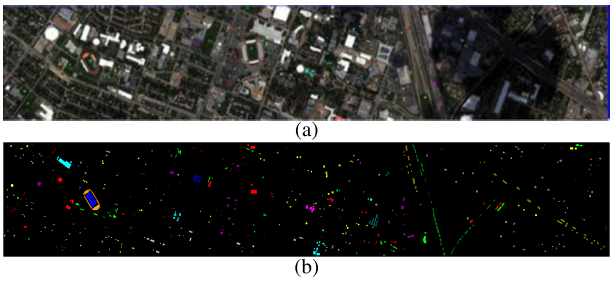
(a) (b)

Fig. 7. (a) Color image of SV dataset. (b) Label map of SV dataset.

TABLE IV
NUMBER OF SAMPLES OF THE HT DATASET

Color	Land-cover-type	Samples
	Healthy grass	1251
	Stressed grass	1254
	Synthetic grass	697
	Trees	1244
	Soil	1242
	Water	325
	Residential	1268
	Commercial	1244
	Road	1252
	Highway	1227
	Railway	1235
	Parking Lot 1	1233
	Parking Lot 2	469
	Tennis Court	428
	Running Track	660
	Total samples	15029

We divide HSI data into $9 \times 9 \times w$. The size of w is input into our model, where w is the number of bands HSI passes through PCA, 128 bands are reserved for IP dataset, and 32 bands are reserved for SV dataset and UP dataset. The learning rate of IP dataset is 0.001, SV dataset is 0.0003, and UP dataset is 0.001. We use Adam to optimize the network and select the Cosine Annealing LR method to reduce the learning rate, where the parameter epoch is set to 20, eta_min is set to 0, the number of training rounds is 10, and the number of training rounds is 100. The batch size of IP dataset is set to 32, and the batch size of SV and UP dataset is 128. The hyperparameter γ in multiclass focal-loss was 2 on the IP dataset and 1 on the other datasets. In addition, the parameters of the model used for comparison are the same as those in the original article. To ensure the accuracy and objectivity of the experiment, all of our training samples were randomly selected, and all the experiments were run for ten times, with each training epoch of 100. For the IP dataset, we randomly select 5% of the total number of samples as the training set, that is, 512 samples, 5% as the verification set, and the remaining marked samples as the test. For SV and UP datasets, we randomly select 1% of the total number of samples as the training set, namely 427 and 541 samples, 1% as the verification set, and the remaining marked samples as the test. We use the overall accuracy (OA), average accuracy (AA), and kappa efficiency as evaluation indicators.



(a) (b)

Fig. 8. (a) Color image of HT dataset. (b) Label map of HT dataset.

C. Result Analysis

To verify the effectiveness of DAFFN in HSI classification, the following four experiments were carried out.

1) Compare the Classification Results of DAFFN With the Current More Advanced Models, Namely DBMA [21], DBDA [22], SSRN [11], fast dense spectral–spatial convolution (FDSSC) [12], lightweight spectral–spatial attention network (LSSAN) [25], spectral–spatial Feature tokenization transformer (SSFTT) [38], MDSCN [18], Attentionbased Adaptive Spectral-Spatial Kernel ResNet (A2S2K-ResNet) [39], and spectral–spatial transformer network (SSTN) [40]: We tested these nine methods on four datasets. Tables V–VIII shows the OA, AA, and Kappa of different algorithms on each figure category of the four datasets. The experimental results obtained by these algorithms are shown in Tables V–VIII, and the classification results are shown in Figs. 9–12. It can be seen from the four tables that our model DAFFN has achieved the best classification results and strong stability on the four datasets. According to the OA, AA, and Kappa in Tables V–VIII, the DAFFN method has increased by about 1%–5% compared with the eight methods with better effect at present. Second, this article adopts 2-D convolution operation to greatly reduce the amount of computation while ensuring the classification accuracy.

SSRN uses a standard residual network. FDSSC adds a dense connection module to the residual network. DBMA

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TABLE V
CLASSIFICATION RESULTS FOR IP DATASET USING 5% OF THE AVAILABLE LABELED DATA

Class	DBMA	DBDA	SSRN	FDSSC	LSSAN	SSFTT	MDSCN	A^2S^2K-ResNet	SSTN	DAFFN
1	89.05	98.56	96.62	**100.00**	90.48	79.06	**100.00**	94.69	80.24	**100.00**
2	93.24	96.63	94.85	98.21	95.82	90.27	98.20	95.67	90.23	**98.49**
3	94.15	95.41	95.45	94.39	93.26	94.55	**95.95**	94.41	89.75	95.06
4	95.96	92.60	93.59	96.04	95.44	91.96	94.15	96.86	95.92	**97.97**
5	94.37	97.82	97.61	95.74	91.17	**99.56**	98.80	98.49	97.18	98.12
6	96.90	98.30	99.14	98.21	99.32	97.25	**99.55**	98.60	98.71	98.63
7	99.26	96.42	98.89	80.00	93.08	95.55	**100.00**	90.06	79.32	**100.00**
8	99.48	**100.00**	98.59	99.81	99.49	99.33	**100.00**	99.86	97.13	**100.00**
9	80.95	**100.00**	**100.00**	80.00	57.78	52.63	80.00	74.52	66.23	78.18
10	91.19	98.78	96.39	**98.86**	96.07	98.26	98.61	92.78	83.19	93.15
11	93.67	97.43	96.66	97.40	97.67	97.46	98.16	96.81	91.38	**98.88**
12	86.06	**98.60**	95.90	97.47	94.88	81.52	97.43	95.62	89.75	94.69
13	99.47	99.52	99.31	**100.00**	99.24	97.94	**100.00**	98.21	97.05	**100.00**
14	97.28	97.80	97.88	**99.49**	98.82	97.91	99.39	98.22	94.76	98.36
15	90.79	94.73	93.22	98.62	95.52	85.55	**100.00**	96.49	87.79	98.86
16	94.05	92.05	94.49	97.17	95.06	92.06	**100.00**	92.98	90.16	97.43
OA	94.95±1.03	96.98±0.63	95.16±1.45	96.43±0.89	96.51±0.45	94.42±0.78	**97.40±0.52**	96.35±0.60	92.34±1.93	97.39±0.37
AA	91.20±2.41	94.50±1.33	94.39±1.76	92.34±1.32	93.30±1.48	95.11±1.32	**96.99±1.36**	94.64±1.63	90.72±3.11	96.28±0.53
Kappa	94.25±1.16	96.95±0.72	94.48±1.66	95.94±0.10	96.02±0.51	87.95±0.88	**97.18±0.59**	95.85±0.68	90.57±2.74	97.02±0.42

TABLE VI
CLASSIFICATION RESULTS FOR UP DATASET USING 1% OF THE AVAILABLE LABELED DATA

class	DBMA	DBDA	SSRN	FDSSC	LSSAN	SSFTT	MDSCN	A^2S^2K-ResNet	SSTN	DAFFN
1	91.67	95.95	97.06	**99.04**	96.97	97.17	98.05	92.49	96.95	99.01
2	98.56	99.54	97.46	99.33	99.75	99.75	99.59	98.06	98.27	**99.89**
3	92.45	95.22	81.92	76.37	75.40	89.60	**95.86**	87.05	92.15	94.49
4	96.26	97.80	99.91	99.50	96.40	86.86	**99.68**	99.32	98.33	98.89
5	98.91	99.62	99.85	99.92	**99.94**	97.51	99.70	99.30	99.41	99.76
6	**99.94**	99.12	97.07	96.52	99.26	98.11	99.23	96.88	97.68	98.83
7	98.27	99.79	94.83	99.31	97.75	98.93	**100.00**	89.13	90.17	**100.00**
8	94.72	90.47	86.26	**97.36**	85.47	83.06	94.10	84.79	88.54	96.34
9	**99.94**	97.56	99.75	99.68	97.31	82.79	97.85	88.65	94.12	97.95
OA%	96.74±0.53	97.62±0.87	95.68±0.78	96.78±2.11	96.48±0.33	96.07±0.65	98.22±0.22	94.89±0.94	95.89±2.13	**98.55±0.19**
AA%	96.75±1.27	97.23±0.32	94.90±1.01	96.34±2.62	94.25±0.65	93.16±1.11	97.58±0.48	92.82±3.96	93.71±3.06	**97.78±0.15**
Kappa	95.67±0.71	96.84±1.16	94.25±1.05	95.76±2.78	95.34±0.44	94.79±0.86	97.64±0.29	93.21±1.25	94.15±2.86	**98.08±0.21**

TABLE VII
CLASSIFICATION RESULTS FOR SV DATASET USING 1% OF THE AVAILABLE LABELED DATA

class	DBMA	DBDA	SSRN	FDSSC	LSSAN	SSFTT	MDSCN	A^2S^2K-ResNet	SSTN	DAFFN
1	**100.00**	61.85	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
2	**100.00**	88.25	91.86	**100.00**	99.96	**100.00**	**100.00**	99.99	**100.00**	**100.00**
3	99.67	99.38	96.37	99.64	**100.00**	98.26	**100.00**	99.92	99.37	**100.00**
4	91.08	96.26	98.75	94.32	99.43	**99.92**	98.61	98.60	98.46	97.76
5	**99.67**	99.11	98.53	99.62	98.52	98.07	99.47	99.73	99.21	99.55
6	99.62	99.94	99.97	99.73	99.96	99.31	**100.00**	99.94	99.95	**100.00**
7	99.98	99.12	99.99	**100.00**	99.83	99.88	**100.00**	99.95	**100.00**	**100.00**
8	93.00	91.07	87.04	94.91	93.45	99.21	98.72	94.69	95.78	**99.89**
9	99.95	99.71	99.46	99.90	99.99	**100.00**	**100.00**	99.89	99.93	**100.00**
10	98.50	99.15	99.33	99.44	95.16	98.48	98.32	98.81	98.45	**99.84**
11	**100.00**	97.82	95.84	**100.00**	99.16	99.52	**100.00**	97.58	98.39	**100.00**
12	99.00	99.91	97.49	99.71	99.92	95.75	**100.00**	99.41	99.62	99.89
13	99.93	99.35	99.40	**100.00**	99.45	97.90	99.78	99.79	99.74	**100.00**
14	93.63	96.77	90.91	96.35	99.33	91.31	99.12	95.41	96.55	**99.27**
15	97.18	81.8	81.43	97.56	93.01	95.88	97.46	91.74	92.63	**99.46**
16	99.69	**100.00**	98.93	98.40	97.90	99.32	**100.00**	99.78	99.90	**100.00**
OA%	97.50±0.47	91.87±7.51	93.13±1.50	98.19±0.20	97.18±0.42	98.34±0.36	98.82±0.29	97.38±0.39	98.01±0.98	**99.49±0.17**
AA%	98.18±0.35	94.35±5.45	95.96±1.19	98.72±0.05	98.44±0.16	98.11±0.55	99.14±0.29	98.41±0.67	98.92±1.53	**99.23±0.16**
Kappa	97.21±0.53	91.01±8.28	92.34±1.69	97.99±0.22	96.87±0.47	98.15±0.41	98.68±0.33	97.09±0.44	98.21±1.76	**99.43±0.19**

and DBDA both add attention modules to the dense connection network. LSSAN is a network model based on spatial spectrum attention mechanism. A concise and efficient spatial attention model is proposed, which can extract the most valuable spatial information while suppressing the interference of invalid pixels. MDSCN extracts spatial local features and spectral local features, respectively. SSFTT is based on the most popular Transformer model. It uses CNN to extract

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TABLE VIII
CLASSIFICATION RESULTS FOR HT DATASET USING 2% OF THE AVAILABLE LABELED DATA

class	DBMA	DBDA	SSRN	FDSSC	LSSAN	SSFTT	MDSCN	A^2S^2K-ResNet	SSTN	DAFFN
1	91.64	91.17	**92.29**	90.81	92.12	91.63	89.93	91.14	87.43	92.21
2	96.92	99.14	98.73	97.71	**99.23**	98.24	95.98	97.51	95.26	98.15
3	99.91	99.86	99.74	**100.00**	99.74	99.78	**100.00**	99.85	98.23	99.86
4	98.09	98.80	**99.19**	99.41	98.91	98.15	94.27	97.96	96.82	95.83
5	96.01	96.02	95.51	**97.77**	96.48	94.95	96.11	94.70	92.19	96.36
6	99.06	98.93	**99.96**	99.52	98.65	99.27	99.50	99.21	98.42	98.82
7	87.54	91.33	89.78	92.57	92.34	90.31	88.12	88.71	88.31	**92.59**
8	93.99	94.48	94.46	95.90	93.96	96.57	90.61	96.60	95.28	**96.98**
9	86.18	90.43	86.96	88.29	89.59	89.91	81.67	84.29	84.92	**92.45**
10	**85.57**	84.48	82.51	83.34	85.11	80.36	78.64	78.18	77.44	83.47
11	93.34	95.02	93.88	**96.92**	94.37	88.24	84.70	87.00	86.17	90.76
12	91.47	93.23	89.85	93.43	**93.54**	88.53	85.03	88.83	86.32	92.96
13	89.52	88.44	**92.16**	88.51	89.26	87.66	79.75	86.09	86.70	89.61
14	97.74	98.76	94.48	96.20	98.79	98.27	**100.00**	96.09	96.21	97.80
15	95.69	96.51	96.85	96.82	96.83	97.89	**99.02**	98.61	97.57	97.01
OA%	90.47±2.44	91.63±1.16	90.66±1.35	91.64±1.77	90.52±0.69	91.02±1.73	89.13±1.49	90.52±1.72	87.49±1.48	**93.39±1.02**
AA%	91.51±1.10	92.44±1.00	91.76±1.27	92.48±1.17	92.35±0.86	93.75±1.32	90.89±1.47	92.32±1.79	89.27±1.35	**94.32±0.84**
Kappa	90.86±1.55	90.11±1.26	90.07±1.46	90.12±1.91	90.67±0.74	90.21±1.26	88.25±1.61	89.75±1.86	86.19±1.83	**92.85±1.10**

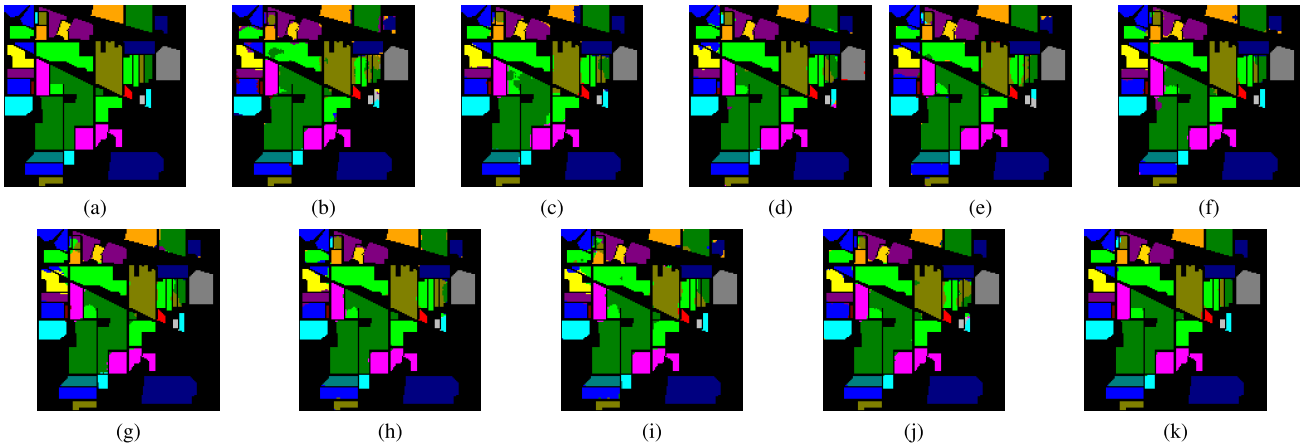


Fig. 9. Classification maps for IP dataset. (a) Ground truth. (b) DBMA. (c) DBDA. (d) SSRN. (e) FDSSC. (f) LSSAN. (g) SSFTT. (h) MDSCN. (i) A^2S^2K-ResNet. (j) SSTN. (k) DAFFN.

shallow spatial spectral features and transformer to extract deep semantic features. SSTN is a transformer network model composed of spectral attention and spatial attention modules, aiming to optimize network parameters and classify HSI under low parameter count and computational complexity. A^2S^2K-ResNet proposes an improved 3-D ResBlocks for extracting spectral spatial features and using an efficient feature recalibration (EFR) mechanism to improve classification performance and capture cross channel dependencies. It can be seen from the table that the classification accuracy of the method using attention mechanism is significantly better than that of the basic CNN structure. Compared with LSSAN, DBDA, DBMA, SSTN, and A^2S^2K-ResNet using attention mechanism, DAFFN has better classification performance, because the model not only focuses on the global attention feature of HSI, but also does not ignore the neighborhood context attention information of HSI, and focuses on the local context feature extraction, Stable and reliable performance is obtained under limited sample conditions. As shown in the table, the OA of DAFFN in the IP dataset is only 0.1% lower than that of the MDSCN model, but the classification accuracy is the highest in both SV and UP dataset. Because the number of samples in the IP dataset is small and the classification difficulty is low. The excessive use of attention mechanisms in DAFFN results in overfitting and over extraction of global local contextual features from IP datasets, resulting in the phenomenon of homospectral foreign objects. As a result, the classification accuracy of IP datasets is slightly lower, while the sample size of UP and SV datasets is about five times that of IP datasets. The features to be extracted are more complex and do not cause overfitting, thus achieving better classification results. Therefore, DAFFN is suitable for datasets with complex and difficult to classify ground features. However, the MDSCN model has a shallow network layer and cannot capture key information for complex datasets, making it suitable for datasets with low classification difficulty. In addition, from the variance data of three datasets OA, AA, and Kappa, we can see that DAFFN is more robust, and the results of ten training sessions are the most stable, which shows that our attention mechanism can accurately extract key features, and is not affected by irrelevant pixel information. It can also be seen from the classification results of the four

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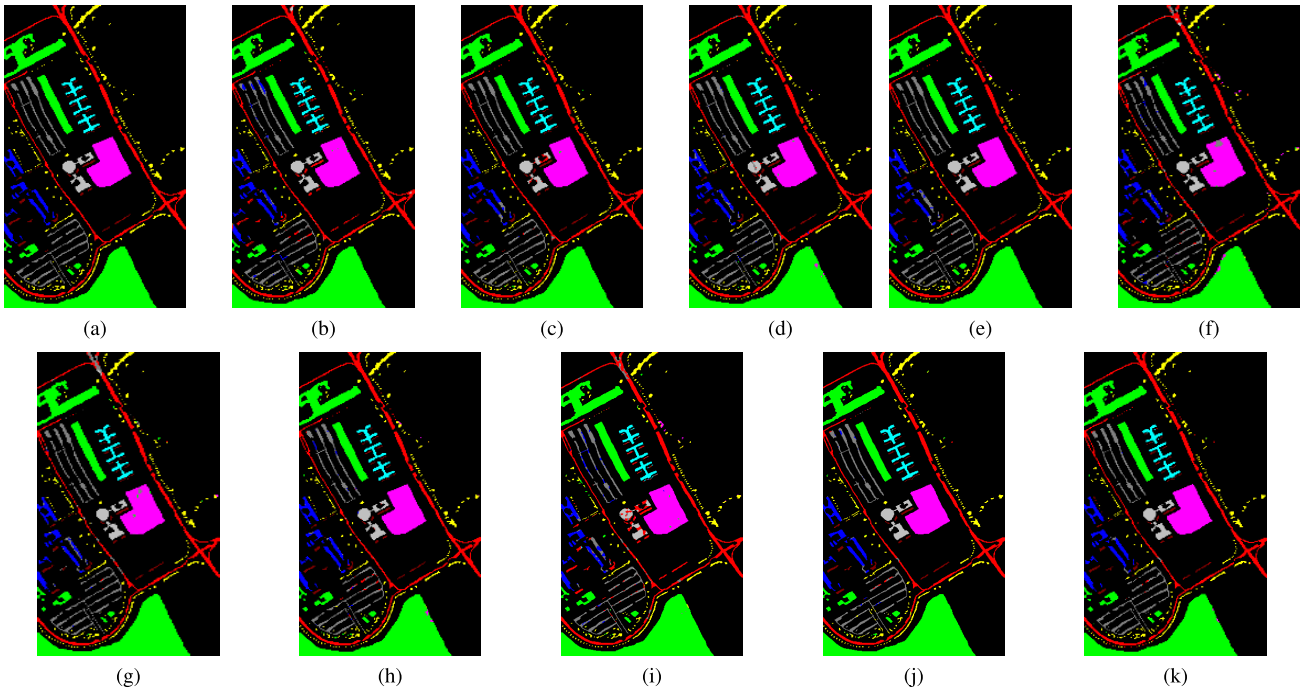


Fig. 10. Classification maps for UP dataset. (a) Ground truth. (b) DBMA. (c) DBDA. (d) SSRN. (e) FDSSC. (f) LSSAN. (g) SSFTT. (h) MDSCN. (i) A^2S^2K-ResNet. (j) SSTN. (k) DAFFN.

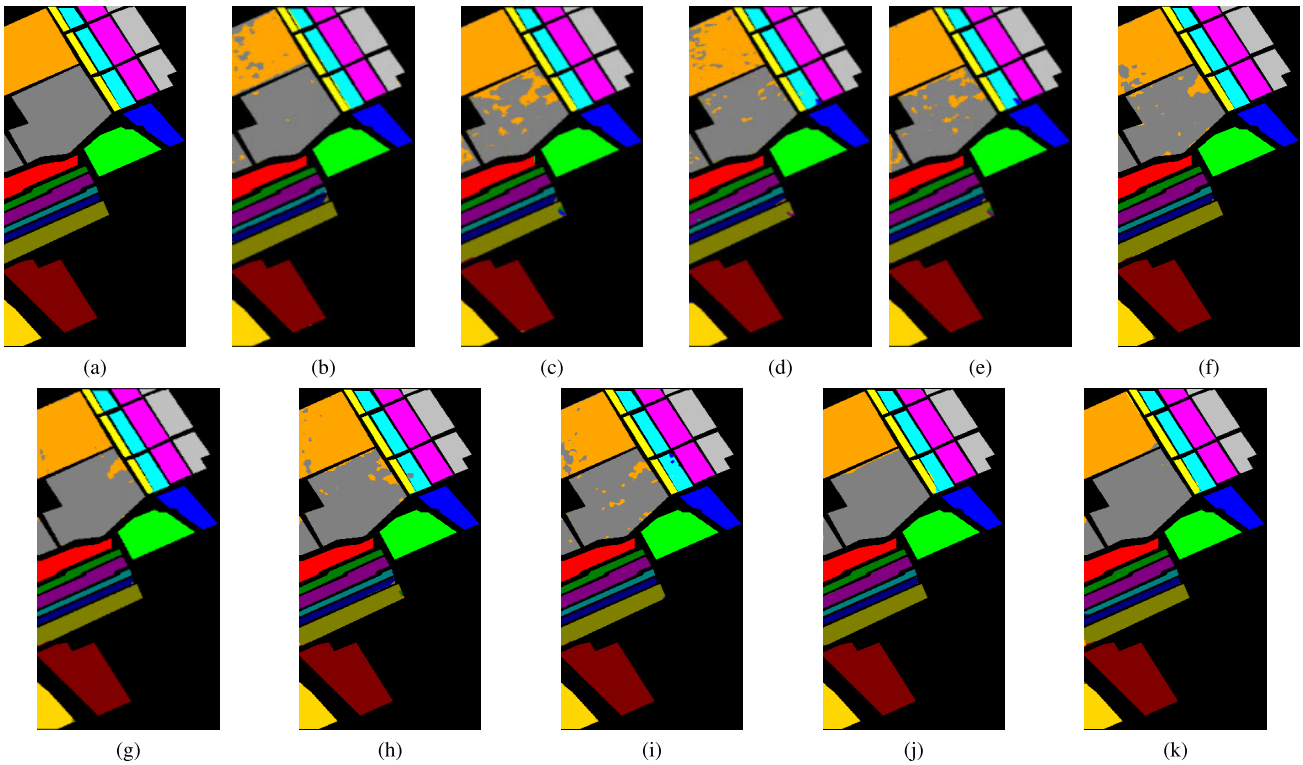


Fig. 11. Classification maps for SV dataset. (a) Ground truth. (b) DBMA. (c) DBDA. (d) SSRN. (e) FDSSC. (f) LSSAN. (g) SSFTT. (h) MDSCN. (i) A^2S^2K-ResNet. (j) SSTN. (k) DAFFN.

datasets in Figs. 9–12 that the error rate of the method in this article is the lowest, and the resulting diagram has the least impurities.

For IP dataset, the ten highest OA of DAFFN reached 98.2%, and the accuracy of classification of ground objects in categories 1, 7, 8, and 13 reached 100%. However, the classification accuracy of class 9 samples is only 78%, lower than that of other models, because the number of samples in this class is only 20, and the number of samples used for training is insufficient, while the network layer of module DAFFN is not as deep as that of other deep learning models, and no distinguishing features are extracted, resulting in the

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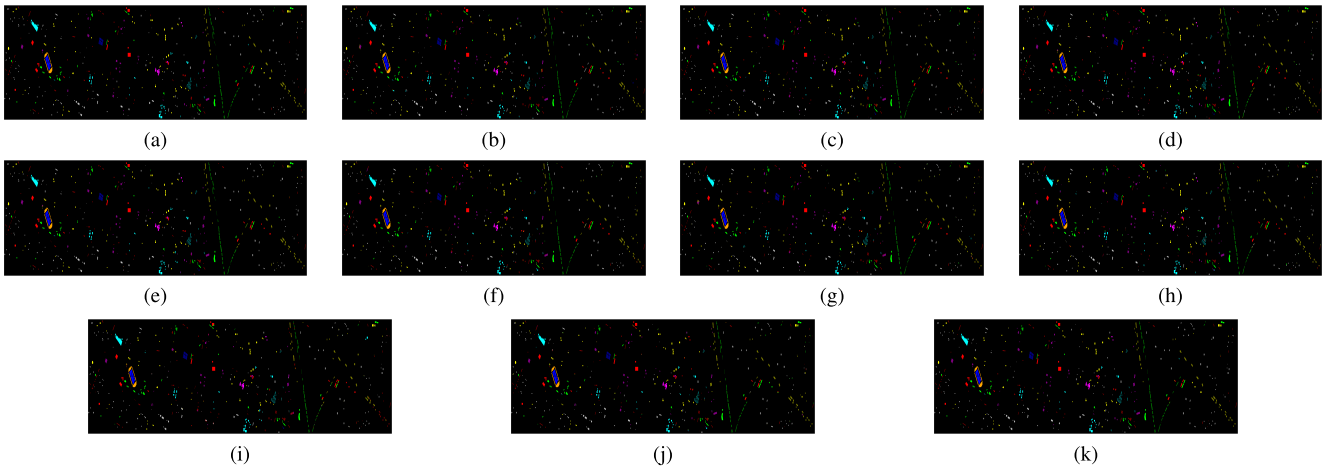


Fig. 12. Classification maps for HT dataset. (a) Ground truth. (b) DBMA. (c) DBDA. (d) SSRN. (e) FDSSC. (f) LSSAN. (g) SSFTT. (h) MDSCN. (i) A^2S^2K-ResNet. (j) SSTN. (k) DAFFN.

TABLE IX
COMPARISONS ON MODEL EFFICIENCY OF DIFFERENT METHODS ON IP, UP, SV, AND HT DATASETS

Dataset	Metric	DBMA	DBDA	SSRN	FDSSC	LSSAN	SSFTT	MDSCN	A^2S^2K-ResNet	SSTN	DAFFN
IP	Train times(s)	276.01	124.34	402.31	982.95	67.07	44.53	27.03	90.73	383.32	44.91
	Test times(s)	16.81	10.35	3.06	7.96	5.21	2.82	1.19	7.79	3.12	2.55
	Params(k)	606.1	382.35	364.17	1259.52	23.54	153.74	254.91	370.86	21.36	307.66
	Flops(GM)	0.24	0.11	0.16	0.18	0.02	0.01	0.02	0.17	0.01	0.02
UP	Train times(s)	95.24	55.96	159.04	788.39	82.66	20.56	14.46	38.28	162.65	22.27
	Test times(s)	42.68	26.12	7.94	21.79	13.75	4.32	2.39	18.56	8.23	6.82
	Params(k)	320.69	202.77	216.54	651.06	43.32	153.74	119.54	220.82	16.81	142.86
	Flops(GM)	0.15	0.07	0.08	0.09	0.01	0.01	0.01	0.09	0.01	0.01
SV	Train times(s)	296.4	133.74	207.94	1033.41	150.21	23.19	13.9	79.37	233.85	27.63
	Test times(s)	98.08	60.91	18.13	46.29	20.83	7.64	2.88	30.66	22.11	9.91
	Params(k)	617.72	589.64	570.31	1249.34	48.92	153.74	140.86	83.67	19.28	445.64
	Flops(GM)	0.25	0.11	0.16	0.18	0.01	0.01	0.01	0.01	0.01	0.01
HT	Train times(s)	78.20	89.33	31.05	63.13	40.28	25.23	24.24	33.31	38.37	28.30
	Test times(s)	11.45	16.36	4.79	14.35	3.85	3.48	3.66	8.10	5.84	4.86
	Params(k)	443.36	280.08	253.55	891.43	38.21	153.74	254.65	283.44	18.16	445.32
	Flops(GM)	0.17	0.08	0.11	0.13	0.01	0.01	0.02	0.12	0.01	0.01

classification effect is not excellent enough. For the UP dataset, the ten highest OA of DAFFN reached 98.88%, of which the accuracy of classification of category 7 ground objects reached 100%. For SV dataset, the ten highest OA of DAFFN reached 99.7%, of which the accuracy of classification of ground objects in categories 1, 2, 3, 6, 7, 9, 11, 13, and 16 reached 100%. For HT dataset, the ten highest OA of DAFFN reached 94.8%, of which the accuracy of classification of ground objects in categories 3, 6, and 14 reached 100%. It can be seen that DAFFN is effective and accurate in feature extraction.

2) Efficiency Analysis: Table IX lists the training time, testing time, model complexity, and computational complexity under different models. It can be seen that although the DAFFN model is not the best result among the four performance indicators, it is at a leading level in terms of time consumption, parameter quantity, and computational cost. DAFFN can achieve the best classification accuracy at the expense of a small amount of performance. Overall, DAFFN has the highest cost-effectiveness.

TABLE X
OA% WITH DIFFERENT PATCH SIZES ON FOUR DATASETS

Patch Size	IP	UP	SV	HT
5×5	95.37	97.10	96.01	90.44
7×7	96.68	97.69	97.67	91.86
9×9	97.39	98.55	99.43	93.39
11×11	97.27	98.42	99.08	91.56
13×13	96.94	98.49	99.23	90.65

3) Impact of the Input Patch Size: Table X shows the impact of changes in batch size on classification accuracy when other parameters are fixed. It can be seen that when the patch size of all four datasets is 9×9, OA reaches the optimal value. Due to different patches, the size of the input image and the size of the output feature map will also vary. A small patch size will cover a limited number of pixels, resulting in the inability to extract effective neighborhood context features, However, if the patch size is too large, accurate semantic information cannot be obtained, which leads to poor convergence of the loss function and affects network training.

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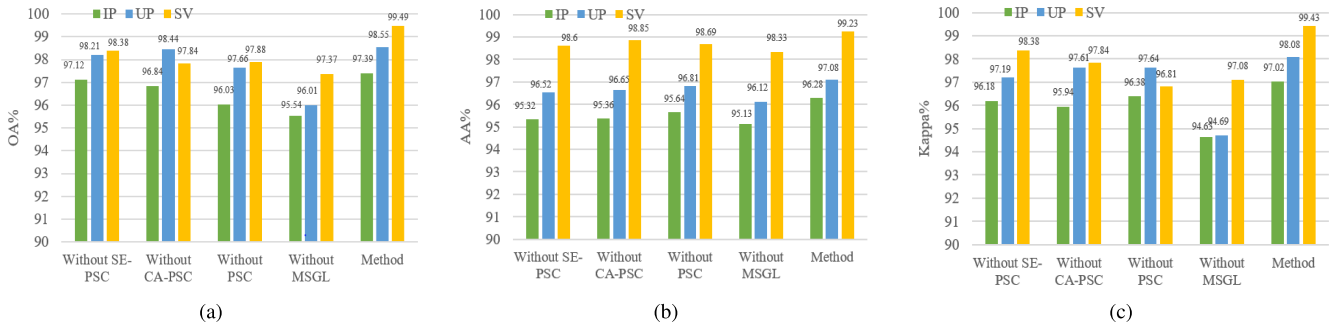


Fig. 13. Effectiveness of different modules. (a) OA. (b) AA. (c) Kappa.

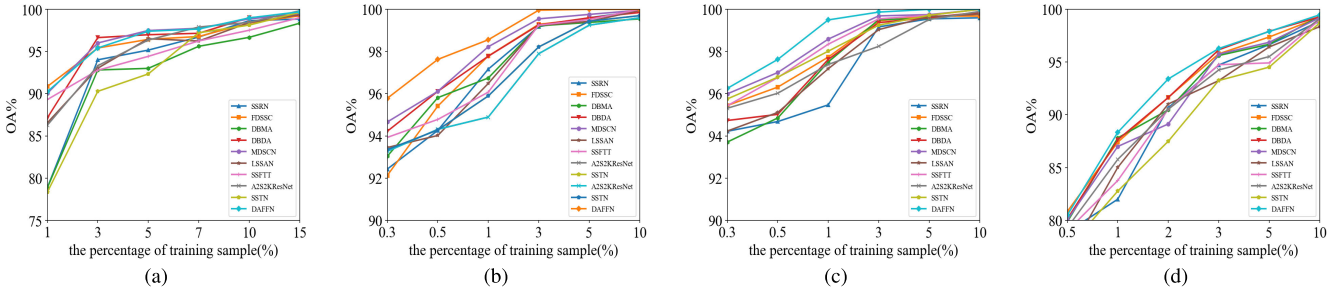


Fig. 14. Classification results of each model under different training set ratios on the four datasets. (a) IP. (b) UP. (c) SV. (d) HT.

4) Verify the Effectiveness of the Module: To verify the effectiveness of the SA module, cross-attention module, PSC module and multiscale feature global–local fusion module in DAFFN, we deleted four modules, respectively, and compared their training results with the model.

As shown in Fig. 13, from the experimental results of OA, the value of OA without adding any module has decreased by more than 1%, and the MSGL module has the greatest impact on the performance of the model, reducing by 1.85%, 2.04%, and 2.12% on the IP, UP, and SV datasets, respectively, which proves the effectiveness of the proposed feature fusion module. Second, compared with DAFFN with position calibration, the OA of the three datasets increased by 1.36%, 0.89%, and 1.61%, respectively. It can be seen that position calibration can help the attention mechanism to extract key features without destroying its position information. The experimental results indicate that when the sample size is severely limited, the use of attention mechanism is more effective, which is conducive to extracting deeper features. The experiment shows that the four modules proposed in feature extraction can effectively improve the classification results.

5) Verify the Accuracy of Model Classification Under Different Proportion of Training Samples: For different proportion of training samples, the classification accuracy of DAFFN and other models is verified. For IP datasets, select 1%, 3%, 5%, 7%, 10%, and 15% as training sets, for UP and SV datasets, select 0.3%, 0.5%, 1%, 3%, 5%, and 10% as training sets, and for HT datasets, select 0.5%, 1%, 2%, 3%, 5%, and 10% as training sets.

It can be seen from the experimental results in Fig. 14 that the classification accuracy on the four datasets is in direct proportion to the number of training samples, and when the number of training samples is enough, it gradually tends to be stable. When the number of training samples in the IP dataset reaches more than 10%, the UP and SV datasets exceeds 3%, and the classification accuracy of each model is close to 100%. The model DAFFN achieves the best results under different training sample numbers.

6) Effectiveness of Feature Extraction: To verify the effectiveness of the method proposed in this article for extracting the upper and lower feature information, we use the gradient-weighted class activation mapping (Grad-CAM) method to visualize the feature map of the HSI classification model. The feature map of each model is extracted from the last convolution layer in the last convolution block. Select one sample with correct classification of all models on three datasets for analysis.

Grad-CAM [41] is a method to judge CNN model by generating visual interpretation based on the classification and positioning technology. Grad-CAM performs backward according to the output vector to obtain the gradient of the feature map and obtain the gradient corresponding to each pixel on each feature map, that is, the gradient map corresponding to the feature map. Then average each gradient graph, that is, the weight of each feature graph, and then sum the weight with the feature graph. Finally, the final class activation graph can be obtained through the ReLu activation function.

It can be seen from Figs. 15–18 that compared with other models, DAFFN model can produce richer and more distinctive feature representations, and can more accurately locate the distribution of target pixels without extending to the background area. DAFFN makes good use of the global–local

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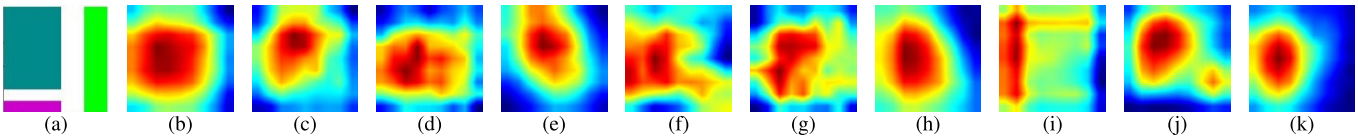


Fig. 15. Grad-CAM maps for IP dataset. (a) Label map. (b) DBMA. (c) DBDA. (d) SSRN. (e) FDSSC. (f) LSSAN. (g) SSFTT. (h) MDSCN. (i) A^2S^2K-ResNet. (j) SSTN. (k) DAFFN.

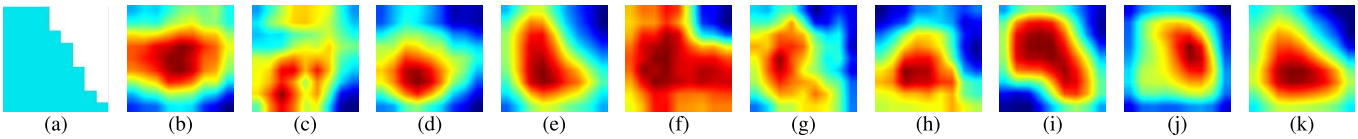


Fig. 16. Grad-CAM maps for UP dataset. (a) Label map. (b) DBMA. (c) DBDA. (d) SSRN. (e) FDSSC. (f) LSSAN. (g) SSFTT. (h) MDSCN. (i) A^2S^2K-ResNet. (j) SSTN. (k) DAFFN.

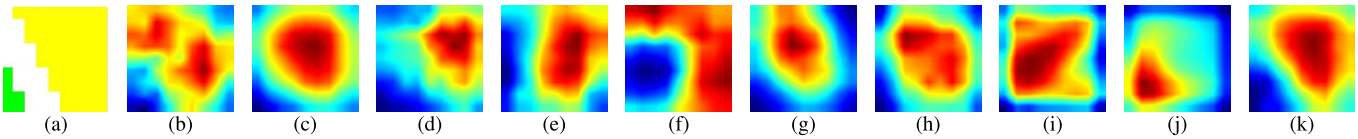


Fig. 17. Grad-CAM maps for SV dataset. (a) Label map. (b) DBMA. (c) DBDA. (d) SSRN. (e) FDSSC. (f) LSSAN. (g) SSFTT. (h) MDSCN. (i) A^2S^2K-ResNet. (j) SSTN. (k) DAFFN.

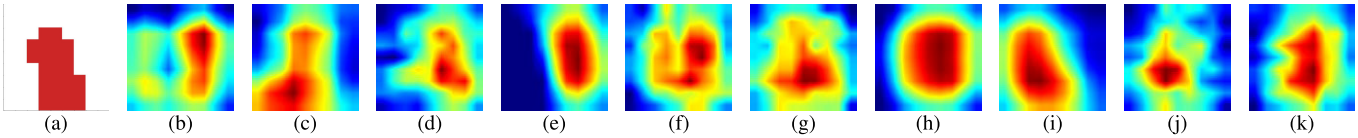


Fig. 18. Grad-CAM maps for HT dataset. (a) Label map. (b) DBMA. (c) DBDA. (d) SSRN. (e) FDSSC. (f) LSSAN. (g) SSFTT. (h) MDSCN. (i) A^2S^2K-ResNet. (j) SSTN. (k) DAFFN.

context feature. Even if the sample feature is small, it can be better limited to the semantic scope and will not be affected by other categories.

IV. Conclusion

In this article, we propose a new feature fusion network model based on dual attention mechanism for HSI classification. This model mainly uses the SA mechanism to extract the global context attention feature and the cross-attention mechanism to extract the local context attention feature. On this basis, we propose a PSC module to compensate for the defect that the attention mechanism can not obtain the position information. This module can be flexibly embedded into any module. In addition, to obtain more effective feature information, we propose an MSGL, which skillfully fuses two parts of features to obtain more robust and accurate recognition results. The experiment shows that DAFFN achieves satisfactory results on four datasets. In the future, we intend to further optimize the model and solve the problem of over-fitting when the number of samples in the dataset is small.

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